MILITARY SPECIFICATION

TROCAR, LICHTWICZ, ANTRUM

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a straight antrum trocar of the Lichtwicz design.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical; General Specifications
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Set-Up.

Military

MIL-P-116	Preservation, Methods of.
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

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STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

DRAWINGS

Defense Personnel Support Center (DPSC)

2144 Trocar, Lichtwicz, Antrum

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNITED STATES OF AMERICA STANDARDS INSTITUTE (U.S.A.S.I.)

Z70.1 American Standard Dimensions of Glass and Metal Luer Tapers for Medical Applications.

(Application for copies should be addressed to the United States of America Standards Institute, Inc., 10 East 40th Street, New York, N.Y. 10016)

3. REQUIREMENTS

3.1 General requirements. The straight antrum trocar shall be in accordance with the applicable requirements of GG-I-526, and as specified herein.

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3.2 Material. The straight antrum trocar shall be fabricated from the following material in accordance with GG-I-526: cannula and stylette shall be class 3, Type 302, 303, or 304; and the hub and tubing connector shall be class 8, alloy number 260, 270 or 360.

3.3 Hardness. After proper stress relieving, the Rockwell hardness of the class 3 material shall be B 75 to B 85.

3.4 Style, design, and dimensions. The style, design, and dimensions of the straight antrum trocar shall conform to the Defense Personnel Support Center (DPSC) Drawing No. 2144 Revision C, dated 19 November 1969, and in accordance with Figure 1 for Luer tip and Figure 3 for female Luer-lock connections, except flanges need not apply, of U.S.A.S.I. Standard Z70.1.

3.5 Finish. The finish shall be mirror , or mirror and satin in accordance with GG-I-526.

3.6 Corrosion resistance. Shall be in accordance with GG-I-526.

3.7 Identification marking. Shall be in accordance with GG-I-526. Instruments of foreign manufacture shall be marked "stainless".

3.8 Workmanship. Shall be in accordance with GG-I-526.

3.9 Luer slip tip. The Luer slip tip shall make a secure, watertight union with a needle hub having a corresponding taper. The tip shall be tested for proper taper and dimensions as specified in 4.4.1, and shall be tested for fluid tightness as specified in 4.4.4.; and when so tested, shall not show leakage exceeding formation of a falling drop within 10 seconds.

3.10 Female Luer-lock connection. Each female Luer-lock connection shall make a secure watertight union with a Luer slip hypodermic syringe and a Luer-lock hypodermic syringe. The female Luer-lock connections shall be tested for depth and diameter as specified in 4.4.2 and for fluid tightness as specified in 4.4.3 and when so tested, shall not show leakage exceeding formation of a falling drop within 10 seconds.

4. QUALITY ASSURANCE PROVISIONS

4.1.2 Certificates of quality. Certificates of quality supplied by the manufacturer of the stainless steel may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the government. The certificate shall include the name of the contractor or supplier, the FSN, the item identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Sampling

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in Table I. The unit of product for examination purposes shall be one trocar.

Table I. Sampling for Examination

Type of Defect	Inspection Level	AQL (Percent Defective)
Major	II	1.0
Minor	II	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in Table II.

Table II. Sampling for Tests

Component (Unit of product)	Characteristic	Reqmt.	Test Procedure	Insp level	AQL (percent defective)
1-Trocar	Hardness	3.3	GG-I-526	S-2	*
1-Trocar	Copper Sulfate	GG-I-526	GG-I-526	S-2	*
1-Trocar	Boil Test	GG-I-526	GG-I-526	S-2	*
1-Trocar	Dimensions	3.4	--------	S-1	*
1-Trocar	Salt Spray Test **	GG-I-526	GG-I-526	S-2	*
1-Trocar	Plating Test **	GG-I-526	GG-I-526	S-2	*
1-Trocar	Luer slip tip	3.9	4.4.1	II	1.0
1-Trocar	Female Luer-lock	3.10	4.4.2	II	1.0
1-Trocar	Fluid tightness	3.9 and 3.10	4.4.3 and 4.4.4	S-4	1.5

* Acceptance number shall be zero for all sample sizes.
** Shall apply for brass material only.

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4.2.3 Raw materials. Four ounces of each lot of each metal used in construction of the antrum trocar shall be selected for chemical anaylsis.

4.2.4 Sample unit usage. Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. Examination shall be in accordance with GG-I-526.

4.3.1 Classification of defects. Examination shall be conducted in accordance with MIL-STD-105 and the classification of defects specified in Table III. Inspection, however, shall not be restricted to the classified possible defects listed in Table III.

Table III. Classification of Defects

Categories	Defects
Critical	None defined.
Major	
101	Trocar not complete.
102	Style and design not as specified.
103	Trocar not free of sharp edges, rough areas, burrs, cracks, grind marks, or overlaps.
Minor	
201	Identification marking not present, correct, complete, or legible.
202	Finish not as specified.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements in accordance with GG-I-526 and as specified herein.

4.4.1 Taper of Luer tip. The male Luer tip shall be tested with the ring gage shown on figure 7 of U.S.A.S.I Standard Z70.1. This gage shall be slipped onto the tip with very light pressure. The tip shall fit the gage within the "GO" - "NO GO" limits of the gage as shown on figure 7 of the U.S.A.S.I Standard. Tips that do not fit the gage within these limits shall be considered not acceptable. In addition, when this ring gage is fitted onto the tip, as specified above, there shall be no movement of the gage on the tip when the gage is pressed lightly from side to side in all directions.

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4.4.2 Diameter and depth. The hub cavity of each female Luer-lock connection shall be tested on the male Luer taper gage in accordance with plug gage of figure 5 of U.S.A.S.I Standard Z70.1. (This gage shall be as shown in the U.S.A.S.I. Standard, except that it shall be constructed with a suitable fluid path to permit performance of the fluid tightness test specified in 4.4.3. The hub shall be slipped onto this gage with very light pressure. The fit of the hub on this gage shall be tested by means of the feeler gage in accordance with figure 5 of U.S.A.S.I. Standard Z70.1. The hub shall fit the plug within the "GO-"NO GO" limits of the feeler gage in accordance with figure 5 of U.S.A.S.I. Standard. Hubs which do not fit the plug within these limits shall be considered not acceptable. In addition, when the hub is fitted onto the plug gage as specified above, there shall be no movement of the hub on the gage when the trocar is pressed lightly from side to side in all directions.

4.4.3 Fluid tightness. With the trocar in a horizontal position, the hub attached by means of a plug gage of figure 5 of U.S.A.S.I. Standard Z70.1 (except that gage shall have a fluid path) to a water pressure source having a pressure gage, the water shall be run through the trocar so as to fill any hub cavity space and lumen, the lumen then sealed by a piece of solid rubber slipped onto the trocar point and a pressure of 45 pounds per square inch applied for 10 seconds.

4.4.4 Fluid tightness for male Luer slip tips. A gage, as shown on figure 8 of the U.S.A.S.I. Standard (except that it shall be constructed with a suitable fluid path), shall be attached to the male Luer slip. The free end of the adapter (tubing end) shall be plugged and the water under pressure of 45 pounds per square inch shall be let in through the gage.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One trocar, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in the following table shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

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Table IV. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	6 units	144 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit preparation. Each needle shall be assembled on a substantial cardboard, plastic, or metal rack, so designed the needle is held securely in place in such a manner that the point cannot be damaged by contact with any part of the container.

5.1.3.2 Unit package. Each unit shall be packaged in accordance with MIL-P-116, method 1A-8.

5.1.3.3 Intermediate package. Intermediate package shall be a carton of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676, except that commercial colors will be acceptable, or PPP-B-636. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3. Closure and strapping shall be as specified in the appendix of the applicable box specification.

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5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commondity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The antrum trocar is used for puncturing antrum to promote drainage and washing out the cavity.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(c) Federal stock number.

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6.3 This specification is intended to cover the following item appearing in the Federal Supply Catalog:
FSN 6515-383-7600, Trocar, Lichtwicz, Antrum. It does not include all types, sizes, or grades as indicated by the title nor all types which are commercially available.

6.4 Supersession data. This specification includes the requirements of Purchase Description No. 3, dated 5 May 1962.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review activities:
Army - MD
Navy - MS
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Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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